Companion Appendix: Recoverable Readout Mismatch in Smooth-to-Spiking EEG Decoding

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Scope. This document records the completed EEG study, supporting analyses, and a subsequent implementation audit. The original primary comparison remains calibrated SAST versus compute-matched ordinary training; the readout analyses were planned secondary outcomes. The study was not preregistered, and the narrative emphasis on recoverable readout mismatch was chosen after inspecting the results. The source-moment diagnostic in Appendix E was added after the main study. A filter-bank common spatial pattern (FBCSP)-style reference was also added after the original run and is reported separately in Appendix F.

The AI4BCI call limits the submission to two pages including references. This companion is kept separate and is not assumed to be an accepted supplement. The main PDF is independently readable.

A Data and neural dynamics

Dataset and partitions. BCI Competition IV 2a, distributed as BNCI2014_001, contains nine participants and four motor-imagery classes (left hand, right hand, feet, tongue) [1]. Each of two sessions has six runs of 48 trials, with 12 trials per class in each run. Only the 22 EEG channels enter the models; the three EOG channels are excluded. All artifact-marked trials are retained. The artifact-free sensitivity evaluates the same fitted models on the unmarked subset.

Partition	Session / runs	Trials/person	Use
Source training	1 / 1–5	240	Weights and preprocessing
Source validation	1 / 6	48	Settings and checkpoints
E1/E2 test	2 / 1–6	288	Conversion and probes
E3 calibration pool	2 / 1	48	Labeled session adaptation
E3 test	2 / 2–6	240	Fixed at every label budget

Preprocessing. MATLAB trial markers precede the cue by two seconds; after converting to zero-based indexing, extraction adds 500 samples and takes 1,000 samples at 250 Hz. A fourth-order Butterworth 4–38 Hz SOS filter is applied forward/backward separately within each run, using odd padding and SciPy's default pad length. Polyphase downsampling by two, with a Kaiser window ($\beta = 5$), produces 22×500 inputs covering 0–4 s after cue onset. Channel means and standard deviations are estimated from the 240 source-training epochs only. The forward/backward filter, padded temporal convolutions, and across-time GroupNorm make this an offline evaluation, distinct from the original competition's causal continuous-output scoring.

Shared spiking-family architecture. An eight-filter temporal convolution (kernel 1×31) feeds spatial mixing across 22 electrodes to 16 channels, average pooling by four, GroupNorm(1), and the first leaky integrate-and-fire (LIF) block. A temporal convolution with kernel 5 maps 16 to 32 channels, followed by GroupNorm(1) and the second LIF block. There are 125 time steps (32 ms each); a linear 32-to-4 head reads their mean activity. The model has 5,900 trainable parameters, no dropout, and no batch running statistics. For each block,

$$u_t = 0.9u_{t-1} + I_t - \theta z_{t-1}, \qquad z_t = \begin{cases} \frac{1}{2} + \arctan(u_t - \theta)/\pi, & \text{smooth,} \\ \mathbf{1}[u_t \geq \theta], & \text{hard.} \end{cases}$$

Initial voltage and activity are zero for each trial and each forward pass. Raw conversion retains $\theta = 1$ and all trained parameters. Changing z_t changes the subsequent reset feedback even though the recurrence equation is unchanged. Direct-hard training uses hard forward activity with the arctan derivative in the backward pass, including reset paths.

B Training, inference, and the original primary result

Ordinary smooth training, same-batch SAM [2], independent-batch SAST [3], and direct-hard training share architecture and initialization for each participant/seed. SAM reuses one minibatch for ascent and descent; SAST uses independently sampled minibatches. Original parameters are restored before each base-optimizer update. The EEGNet-8,2 reference [5] is an independent PyTorch implementation with a 32-sample temporal kernel, depthwise spatial filters, separable temporal convolution, batch normalization, ELU, pooling by 4/8, and dropout 0.5. Spatial and classifier max-norm limits are 1 and 0.25. It has 2,164 parameters and uses PyTorch initialization and the common optimizer, rather than reproducing the paper's full training pipeline.

Optimization and selection. AdamW uses batches of 32 sampled with replacement, 100 epochs of eight updates (200 epochs for matched ordinary), and cosine decay to 1% of the initial rate. Checkpoints are evaluated at 20 equally spaced positions in each method's budget, selected by native-forward validation BA and then loss; all prescribed updates run. There is no gradient clipping, mixed precision, or TF32. Each participant/method has four tuning candidates on pilot seed 1000: learning rate $\{0.0003, 0.001\}$ crossed with radius $\{0.05, 0.2\}$ at weight decay 0.0001 for SAM/SAST, or weight decay $\{0.0001, 0.001\}$ for the other methods. Ties between candidates use validation loss and then candidate order. Main seeds are 0–4. All 216 pilot fits and 270 main fits completed before target-session evaluation.

Table A1: Recorded main-fit compute on NVIDIA L4; means over 45 fits. Time covers training and its validation, excluding data preparation, search, and post-training analysis.

Method	Optimizer updates	Gradient evaluations	Mean seconds
Ordinary	800	800	69.8
Ordinary 2×	1600	1600	139.2
SAM	800	1600	139.4
SAST	800	1600	139.7
Direct hard	800	800	84.7
EEGNet	800	800	3.8

Threshold calibration. One threshold shared by both blocks is selected from $\{0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 3\}$ using source-validation BA. Ties favor the closest value to 1, then the smaller value. Calibration changes both firing threshold and reset magnitude. Direct-hard receives the same calibration. The maximum grid value is selected in 17/45 ordinary, 18/45 matched, 12/45 SAM, and 14/45 SAST runs; conclusions concern this finite grid, not optimal possible calibration.

Inference. BA is the unweighted mean of four class recalls. Five seeds are averaged within each participant before computing means and 10,000 percentile bootstrap resamples (random seed 20261012). Contrasts are paired within participant; the sampling unit is the participant ($n = 9$). Intervals are not corrected for the multiple secondary comparisons. The primary effect, calibrated SAST minus calibrated matched ordinary, is -0.42 pp, 95% CI $[-2.88, 2.67]$. The exact 2^9 sign-flip sensitivity test gives $p = 0.8164$, conditional on sign exchangeability. Restricting evaluation to artifact-free trials gives -0.32 pp $[-2.87, 2.87]$.

Participant	1	2	3	4	5	6	7	8	9
Difference (pp)	$+9.72$	$+1.87$	-0.21	$+1.25$	-2.50	-4.51	-2.64	-1.04	-5.69

The interval includes meaningful effects in either direction; it does not establish equivalence. Neither this comparison nor the training algorithm alone identifies a causal effect of parameter-space flatness.

C Conversion loss, readout recovery, and representation alignment

Matched probe construction. Each mode and layer receives a separate ridge classifier on time-averaged features. Source-training feature means and standard deviations (floored at 10^{-6}) define the scaler. A centered least-squares fit to one-hot labels includes an unpenalized intercept, with penalty $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ chosen by source-validation BA; ties favor larger α. All modes use the same 240 fitting and 48 validation trials. The 288 target trials are used only for evaluation. The final-layer raw-versus-smooth probe contrast and its paired interval were calculated during the final analysis from the planned probe outputs.

Table A2: Within-checkpoint conversion loss and degenerate predictions. Positive losses mean that smooth evaluation is more accurate. Collapse is defined as predicting one class for all 288 target trials, out of 45 fits.

	Smooth minus spiking BA, pp [95% CI]		Single-class fits	
Method	Raw	Calibrated	Raw	Calibrated
Ordinary	15.97 [8.86, 23.43]	8.54 [4.96, 12.26]	20	4
Ordinary 2×	17.21 [10.22, 24.34]	9.08 [4.77, 13.57]	19	4
SAM	12.08 [5.25, 19.37]	5.27 [1.99, 8.97]	12	5
SAST	13.56 [7.13, 20.40]	6.43 [3.29, 9.87]	13	3

Table A3: Final-layer linear probes: BA (%) and paired effects (pp). Recovery is raw-probe minus raw-original-head accuracy. All refitting uses source-session labels only.

Method	Smooth probe	Raw probe	Recovery [95% CI]	Raw − smooth [95% CI]
Ordinary	43.23	41.13	15.86 [9.23, 22.93]	-2.09 [-3.09, -1.08]
Ordinary 2×	43.67	42.37	17.04 [10.57, 23.46]	-1.30 [-2.35, -0.33]
SAM	40.93	38.69	13.16 [6.28, 20.56]	-2.25 [-3.02, -1.55]
SAST	41.91	40.53	14.62 [8.26, 21.37]	-1.37 [-2.03, -0.71]

Refitting smooth features also improves accuracy: ordinary $+1.98$ pp, matched ordinary $+1.13$ pp, SAM $+3.32$ pp, and SAST $+2.44$ pp. This control limits how much of raw-spike recovery can be specifically attributed to repairing conversion. The remaining negative raw-minus-smooth probe contrasts support a residual loss of linear decodability under the chosen readout family; they do not establish loss of all task information. Calibrated-spike final-layer probe means are 40.27%, 41.77%, 38.94%, and 40.59% in the same method order. Direct-hard raw features reach 44.30% with a ridge head; EEGNet features reach 54.88%.

Geometric similarity. Linear centered kernel alignment (CKA) [6] compares paired smooth and hard features on the target trials, centered across trials. For centered feature matrices X, Y,

$$\mathrm{CKA}(X, Y) = \frac{\|X^\top Y\|_F^2}{\|X^\top X\|_F \, \|Y^\top Y\|_F}.$$

A degenerate denominator is left undefined. Final-layer CKA can remain high despite collapsed original-head predictions.

	Layer 1 CKA		Layer 2 CKA	
Method	Raw	Calibrated	Raw	Calibrated
Ordinary	0.992	0.981	0.903	0.890
Ordinary 2×	0.994	0.985	0.876	0.879
SAM	0.947	0.935	0.801	0.823
SAST	0.986	0.983	0.900	0.914

SAST final-layer raw CKA is 0.900 [0.865, 0.928], and calibrated CKA is 0.914 [0.883, 0.935]. These are descriptive geometric measures; high CKA does not establish preserved decisions, identical dynamics, or equivalence.

D Limited-label target-session adaptation

E3 is a planned secondary experiment with a different test partition from E1/E2. All of target run 1 is reserved for calibration at every budget, including zero. Nested calibration sets take the first $k = 0, 4, 8, 12$ trials per class in recording order, giving 0, 16, 32, or 48 labels. Target runs 2–6 (240 trials) form the fixed test set. No unused trial from the calibration run enters the test set.

Readout only. The encoder and source-selected threshold are frozen. The ridge head is refitted on source-training plus calibration trials with equal trial weights; feature scaling and α remain fixed from the source-only fit. The zero-label baseline is already a source-fitted ridge head, rather than the original trained classifier. Accordingly, gains below measure the extra benefit of target labels after source-only refitting.

Table A4: Readout adaptation on a fixed target test set. Entries are mean BA (%). The final column is the paired 48-minus-0-label gain, pp [95% CI]. The spiking modes use source-selected thresholds.

Method	Mode	0	16	32	48	Gain [95% CI]
Ordinary	Smooth	43.08	43.51	43.72	44.33	1.25 [0.45, 2.12]
Ordinary	Spikes	40.19	40.49	41.09	41.49	1.30 [0.42, 2.37]
Ordinary 2×	Smooth	43.38	43.75	44.19	44.19	0.81 [-0.09, 1.94]
Ordinary 2×	Spikes	41.54	41.76	42.30	42.43	0.89 [0.17, 1.62]
SAM	Smooth	40.88	41.23	41.45	41.68	0.80 [-0.07, 1.60]
SAM	Spikes	38.90	39.11	39.38	39.35	0.45 [-0.20, 1.11]
SAST	Smooth	41.96	42.45	42.75	43.11	1.15 [0.48, 1.81]
SAST	Spikes	40.47	40.97	41.06	41.46	0.99 [0.43, 1.54]
Direct hard	Spikes	44.44	44.90	45.11	45.24	0.81 [0.32, 1.28]
EEGNet	Native	53.88	54.15	54.63	55.50	1.62 [0.94, 2.42]

Threshold only. All parameters, including the original head, are fixed. At zero labels the source-selected threshold is used; at positive budgets the same eight-candidate grid is selected by calibration-trial BA, with ties closest to the source threshold and then toward the smaller value. These zero-label baselines differ from the readout branch.

Table A5: Threshold-only adaptation, mean BA (%), with the same fixed test set and label budgets. The final column is the paired 48-minus-0-label gain.

Method	0	16	32	48	Gain, pp [95% CI]
Ordinary	32.72	32.30	32.06	32.31	-0.42 [-1.94, 0.60]
Ordinary 2×	33.44	33.31	32.67	33.45	0.01 [-0.42, 0.46]
SAM	32.18	32.34	32.48	33.00	0.82 [-0.14, 2.04]
SAST	33.02	32.46	32.78	33.26	0.24 [-0.74, 1.14]
Direct hard	42.97	40.13	41.22	42.44	-0.54 [-1.30, 0.46]

For calibrated SAST, 48 target labels add 0.99 pp $[0.43, 1.54]$ after source-only ridge refitting, compared with 0.24 pp $[-0.74, 1.14]$ for threshold-only adaptation. These increments are smaller than the source-only readout recovery in E2, but cannot be directly subtracted from E2 because the target test partitions differ. Calibration budgets, interventions, and secondary contrasts were not optimized on the held-out test trials. The approach assumes retained access to the source training data.

E Implementation audit and exploratory source alignment

Audit coverage. The archived engine, local source, and engine embedded in the notebook have identical SHA-256 hashes. All 675 E1 participant/seed/mode BA values and predicted-class counts were recomputed from archived confusion matrices and agreed to numerical precision. For participant 1, seed 0, three checkpoints (SAST, matched ordinary, direct-hard) were replayed on CPU. All predictions matched across eight applicable mode evaluations; maximum absolute feature differences were below 6.6×10^{-7}.

The sampled audit also reproduced training-only input normalization and all 24 source-validation threshold-grid scores, checked trial-state resets and inference-batch invariance, and confirmed that conversion left weights unchanged. An independent scalar recurrence agreed with the implemented smooth and hard dynamics. This audit found no discrepancy explaining the observed collapse within its tested scope. It was not an exhaustive replay of all 270 main checkpoints and did not retrain models.

Exploratory alignment. To test whether simple feature offsets and scales could explain part of the mismatch, the original classifier was held fixed for the SAST and matched-ordinary checkpoints above. Let μ_r, σ_r and μ_s, σ_s be per-feature source-training moments under raw and smooth dynamics. Mean-only correction uses $z'_r = z_r - \mu_r + \mu_s$. Mean-and-scale correction uses

$$z'_r = (z_r - \mu_r) \odot \frac{\sigma_s}{\max(\sigma_r, 10^{-6})} + \mu_s.$$

Moments use the 240 source-training trials without their labels; no target labels or target moments enter the transform. These diagnostics were added after inspecting study outcomes and evaluated on a convenience sample of one participant and one seed, rather than across the full benchmark.

Table A6: Exploratory participant-1/seed-0 BA (%) with the original head. These two checkpoints are illustrative and are not an aggregate result.

Method	Raw	Mean only	Mean and scale	Smooth reference
SAST	25.35	38.89	47.57	53.47
Matched ordinary	25.00	38.54	45.49	55.56

For this SAST checkpoint, the mean across-feature trial standard deviation falls from 0.257 in smooth mode to 0.088 in raw mode; 41.2% of raw time-averaged feature entries are zero. Spiking activity is therefore not universally absent. The alignment response is consistent with a readout mismatch involving feature moments, but does not establish a general repair method or isolate a causal mechanism. A full participant/seed evaluation of fixed source-moment correction remains unrun.

Interpretive limits. The evidence concerns one dataset, one convolutional/LIF architecture, source-selected settings, and an offline participant-specific session shift. Probe success does not prove identical representations; residual probe differences do not identify all information loss. Artifact-free evaluation cannot rule out every ocular or muscle contribution. EEGNet and the FBCSP-style reference remain more accurate than the converted models under these implementations. No hardware energy, real-time latency, cross-participant transfer, or clinical efficacy was measured.

F Post-study classical decoding reference

Scope and method. An additional FBCSP-style reference follows the filter-bank/common spatial pattern (CSP) structure of Ang et al. [7], adapted to the existing offline front end. The source-training runs 1–5, source-validation run 6, all-session-2 test set, cue epochs, channel scaling, and trial retention match E1/E2. Nine fourth-order Butterworth bands are applied forward/backward within each 125 Hz epoch: 4–8, 8–12, 12–16, 16–20, 20–24, 24–28, 28–32, 32–36, and 34–38 Hz. The last band overlaps the preceding band to stay within 4–38 Hz.

Each band has four one-vs-rest CSP models, with four extreme components per model, covariance shrinkage 0.1, full rank, and no trace normalization. The 144 log-mean-square-power features are standardized with source-training statistics. Source-training mutual information (three neighbors; random seed 20261012) selects 8, 16, 32, or 64 features. Multiclass linear discriminant analysis (LDA) uses the least-squares solver and automatic covariance shrinkage. Validation BA selects a candidate, with ties resolved by validation log loss and then candidate order. The selected model is not refitted on validation trials. All nine source choices are locked before session 2 evaluation. This is one deterministic baseline per participant; EEGNet retains its original five-seed participant mean. The original FBCSP competition method used causal Chebyshev filtering and a Naive Bayesian Parzen Window classifier, so this reference is an adaptation rather than an exact reproduction.

Results. FBCSP-style BA is 61.81% [52.93, 70.06], compared with 54.14% [43.83, 64.21] for EEGNet. The paired participant-level difference is +7.66 pp [3.67, 11.91]. Intervals use the same 10,000 participant bootstrap resamples and seed as the original analysis. This comparison was added after the original study; its intervals are descriptive, not corrected for multiple comparisons, and do not replace the primary SAST contrast.

Participant	Features	FBCSP BA (%)	EEGNet BA (%)	Difference (pp)
1	32	75.00	67.78	+7.22
2	32	44.10	42.57	+1.53
3	32	72.57	70.49	+2.08
4	64	59.72	41.25	+18.47
5	8	45.14	31.11	+14.03
6	64	42.71	34.72	+7.99
7	64	73.26	58.06	+15.21
8	64	68.75	66.94	+1.81
9	32	75.00	74.38	+0.63

Provenance. The completed baseline run is `FBCSP_LDA_fb436529d768932f`. It used Python 3.13.16, NumPy 2.1.3, SciPy 1.16.3, pandas 2.2.3, MNE 1.10.2, and scikit-learn 1.7.2. The engine SHA-256 is:

```
4c694da2ab6d74d1dcb8a7428b5d240b
f6572c2a71aea9d91f45403645722cd3
```

The archived engine matches the locally tested implementation. An independent export audit recomputed all nine BA scores and paired intervals from 2,592 saved test predictions and checked model, selection, reference, and split records. Model files were hash-checked; the nine fits were not independently retrained for this audit.

F.1 Post-study EEGNet source-validation audit

The original EEGNet reference was audited without using target labels or changing the training procedure. All 45 main fits and 36 tuning fits completed 800 updates and 20 source-validation evaluations. Recomputed checkpoint choices and all nine optimizer-grid winners agree with the archive. Finite weights, spatial/head max-norm constraints, and BatchNorm update counts also pass. Across main fits, the median selected epoch is 80; 20/45 select epoch 90 or later, including 16/45 at epoch 100. Minimum validation loss occurs at epoch 100 in 20/45 fits, and 13/45 improve their best validation accuracy in the final 25 epochs. These histories do not establish convergence or that longer training would improve target-session performance. EEGNet is a reference under the common study budget and four-candidate source search, not a tuned benchmark ceiling. The audit script, all source learning curves, and input hashes accompany the release.

G Post-study second-architecture readout check

This exploratory check asks whether readout recovery also occurs in a structurally different spiking encoder. It uses the same nine participants, source/target session split, artifact retention, and offline preprocessing. It is not an independent dataset replication and does not repeat the SAST comparison.

Architecture and fixed protocol. A 63-sample temporal convolution has eight filters shared across electrodes. Depthwise spatial filtering learns two electrode combinations per temporal filter, producing 16 channels. Average pooling by eight and GroupNorm(1) precede one LIF block (leak 0.9, threshold 1, zero initial state, subtractive previous-activity reset). Four temporal-bin averages form a 64-dimensional feature vector and a linear four-class head. The encoder has 1,164 trainable parameters. Linear spatial filtering is evaluated before temporal filtering to reduce intermediates; a reference-value and gradient check verifies the equivalent filter ordering. This differs from the original encoder's two LIF blocks, full spatial mixing, and one temporal average per final channel.

Only ordinary smooth training is used. The original 800-update AdamW budget, batch sampling, cosine schedule, four learning-rate/weight-decay candidates, pilot seed 1000, and best-checkpoint rule are retained. Five main seeds per participant use the source-selected setting: 36 tuning fits and 45 main fits. Matched ridge heads are fitted separately to smooth and raw-spike features using the original source-training data and seven-value validation penalty grid. All participants' source fits and readout selections are locked before loading target sessions; file hashes are checked again after evaluation. Encoders remain frozen. No target labels tune the architecture, optimizer, checkpoint, or readout.

The design was fixed before this check's target evaluation, after the original study and its classical reference had been inspected. It was not preregistered. Seeds are averaged within participant before the same 10,000-resample participant bootstrap. Smooth-refit recovery is retained as the control for generic refitting benefits; the excess raw recovery is a paired difference of recovery gains.

Raw original-head BA is 27.44 [26.25, 28.78]%; source-only raw-spike ridge BA is 37.55 [32.59, 42.61]%, giving a paired gain of 10.11 [5.86, 14.29] pp. Smooth original-head BA is 36.44 [31.60, 41.57]% and smooth-ridge BA is 39.46 [34.98, 43.85]%, with gain 3.02 [0.70, 5.42] pp. The raw-minus-smooth matched-ridge contrast is -1.91 [-3.14, -0.61] pp; excess raw recovery over smooth recovery is 7.09 [2.01, 12.21] pp. Intervals are paired participant-bootstrap intervals. These estimates describe this architecture check; they do not add independent participants or establish generality across datasets.

Table A7: Second architecture: session 2 balanced accuracy (%), averaged over five seeds within each participant. Original and ridge heads share the same frozen encoder.

Participant	Smooth original	Smooth ridge	Raw original	Raw ridge
1	40.56	43.19	29.38	43.40
2	26.11	28.89	26.11	25.07
3	51.67	51.25	31.67	51.88
4	30.49	40.83	27.85	36.25
5	26.11	32.78	24.79	28.96
6	34.86	31.46	27.08	32.29
7	35.90	38.89	27.99	35.90
8	41.18	43.82	25.62	41.74
9	41.11	44.03	26.46	42.43

H Reproducibility, provenance, and declarations

The original run, `BNCI2014_001_ccebe0c6749374ab`, completed on 13 October 2026 using NVIDIA L4, Python 3.13.15, and PyTorch 2.11.0+cu130. The CPU audit and second architecture use Python 3.12.12 and PyTorch 2.14.0; full dependency versions are archived with each run. The experiment-engine SHA-256 is:

`42ba021e26724e3e0d18ec77d1df765cc4380`
`92cf487c2eb1b94b8d5a56aacde`

Public reproduction package:
`https://github.com/PluvioXO/ai4bci2026-readout`.
Full run archives are versioned release assets. The package contains:

- `ai4bci_all_experiments.ipynb`: the standalone original experiment runner with its exact archived engine.
- `results/` and `build_paper_results.py`: per-seed outputs, training costs, and regeneration of paper figures, tables, and paired intervals.
- `conversion_audit/`: numerical checks, sampled forward replay, and explicitly exploratory alignment results.
- `ai4bci_fbcsp_baseline.ipynb` and `results/fbcsp_baseline/`: the classical runner, source selections, models, predictions, and export audit.
- `ai4bci_readout_replication.ipynb` and `results/architecture_replication/`: the standalone second-architecture runner and its participant/seed results, intervals, fixed protocol, and export audit.
- `code/`: both neural engines, source-history and prediction audits, and preparation of the three-checkpoint conversion replay.

Full archives retain source configuration, input checksums, splits, histories, selected weights, features, predictions, and calibration identifiers. The main paper compiles independently with standard LaTeX and PGFPlots. Dataset recordings and prepared EEG epochs are not redistributed.

Relationship to other work. SAST was introduced in the author's event-based and on-sensor spiking-network preprints [3, 4]. The present paper applies SAST to EEG motor imagery and adds a planned source-only readout analysis on frozen spiking features. The related UniReps manuscript, *On Representational Fidelity Across Neural Dynamics: Does Flatness Preserve Continuous-to-Spiking Computation?*, is submitted but unpublished and addresses a separate question.

Data and ethics. This is secondary analysis of publicly distributed EEG recordings; no new participants were recruited and no additional recordings were collected. The dataset is obtained from its original distribution and is cited in the main paper. No institutional approval or exemption number is asserted here. The applicable institutional secondary-analysis determination remains unconfirmed.

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